



KRN - LOG - 2026 - 09

Kernellum Build Log

Curated technical milestones from the first public compiler release through the current physical-hardware threshold.



01 / RELEASE

17 September 2026 - v0.1 release

Commit: 63e4a54d4c046862d90a2d7ff98e3791a194160e

Kernellum Compiler v0.1 established the first public end-to-end path: model training, INT8 quantization, constraint-driven architecture search, independent cycle execution, generated SystemVerilog, weights, vectors, and verification collateral.

02 / RTL

17 September 2026 - synthesis safety

Commits: 7a5913072338988f7bc4382bca5489a32d1ba92e; af74d7a9d9ee243df63ac293582d2efd7bf73381

Generated RTL was patched for synthesis safety, including combinational-loop initialization. Synthesis evidence was not treated as valid until these issues were fixed.

03 / REPORT

17 September 2026 - TR-001 and v0.2 alpha

Commit: b4ce966793a95c5cfbb6bda0ed20eb0fb6e91ffb

TR-001 froze the v0.1 evidence record. In parallel, v0.2 alpha introduced a deliberately narrow ONNX Gemm/ReLU front-end and explicit hardware IR.

04 / EDA

17 September 2026 - ONNX-generated RTL + ECP5 mapping

Commit: 0bf4867c1477b04c6b6342241b9a2cf309cbf7f3

CI began building a real ONNX model through the compiler path and simulating/synthesizing its generated RTL. ECP5 family synthesis evidence was exposed publicly.

05 / RECORD

18 September 2026 - public evidence surfaces

The website gained a curated Build Log, claim-to-artifact Evidence Explorer, and FPGA Bring-up Tracker. These surfaces do not add technical evidence; they make the existing evidence easier to inspect and the missing evidence harder to obscure.

06 / NEXT

Current threshold

The next substantive milestone is physical FPGA evidence: named board, real package/constraints, place-and-route, achieved timing, programmed bitstream, measured inference latency, and measured power/energy.

Tracking issue: <https://github.com/sushxnthd/kernellum/issues/1>

07 / REFERENCES

Commit index



SHORT SHA	MILESTONE	DATE
63e4a54	Release Kernellum Compiler v0.1	2026-09-17
7a59130	Patch generated RTL for synthesis safety	2026-09-17
af74d7a	Fix combinational loop initialization	2026-09-17
b4ce966	Publish TR-001 and start v0.2 alpha	2026-09-17
0bf4867	Verify ONNX-generated RTL and expose ECP5 evidence	2026-09-17
2707f2f	Evidence/log/hardware integration phase	2026-09-18